PHYS30762 Object Oriented Programming in C++

Assignment 5 Report

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Contents

1 Program summary **1**

2 References **2**

Note to markers: For version history please see the README file for different branches on the Github repository for assignment 5, as this contains planning and content for this report.

1. Program summary

This program is based around a `Lepton` class, which is an abstract base class from which `Electron`, `Muon`, `Tau`, `Neutrino` are derived classes. The `Lepton` class contains a unique pointer to a `FourMomentum` object, which is an object of a class containing energy, x momentum, y momentum, z momentum, and rest mass, all measured in MeV.

2. References

The following sources were used in the research for this assignment.

[1] https://www.geeksforgeeks.org/cpp/pure-virtual-functions-and-abstract-classes/

[2] https://en.cppreference.com/w/cpp/language/rule_of_three.html

[3] https://www.geeksforgeeks.org/cpp/understanding-constexper-specifier-in-cpp/

[4] https://www.geeksforgeeks.org/cpp/sqrt-sqrtl-sqrtf-cpp/

[5] https://www.geeksforgeeks.org/cpp/enum-classes-in-c-and-their-advantage-over-enum-datatype/

[6] https://www.geeksforgeeks.org/cpp/range-based-loop-c/

[7] https://en.wikipedia.org/wiki/Particle_decay

[8] https://physics.stackexchange.com/questions/282245/muon-decay-mode

[9] https://en.wikipedia.org/wiki/Muon

[10] https://en.wikipedia.org/wiki/Tau_(particle)

[11] https://tikz.net/sm_decay_matrix/

[12] https://www.geeksforgeeks.org/cpp/stduniform_real_-distribution-class-in-c-with-examples/

[13] https://www.w3schools.com/cpp/cpp_switch.asp

[14] https://www.geeksforgeeks.org/cpp/can-constructor-private-cpp/

[15] https://www.w3schools.com/cpp/cpp_constructors_overloading.asp

[16] https://www.geeksforgeeks.org/system-design/factory-method-pattern-c-design-patterns/

[17] https://pdg.lbl.gov/2020/AtomicNuclearProperties/HTML/tungsten_W.html

[18] https://en.wikipedia.org/wiki/Particle_shower

[19] https://indico.capa.unizar.es/event/23/sessions/48/attachments/287/393/COST-TS-Belgrade-Course5-Tutorial.pdf